

# Lab Assignment 02



Inspiring Excellence

|                  |                                                   |
|------------------|---------------------------------------------------|
| Course Code:     | CSE111                                            |
| Course Title:    | Programming Language II                           |
| Topic:           | OOP Basics, Instance Variable and Instance Method |
| Number of Tasks: | 12 (Classwork: 06, Homework: 06)                  |

***[Submit all the Coding Tasks (Homework: Task 1 to 4) in the Google Form shared on buX before the next lab. Submit the Tracing Tasks (Homework: Task 5 to 6) handwritten to your Lab Instructors at the beginning of the lab]***

[You are not allowed to change the driver codes of any of the tasks]

## CLASSWORK

### Task 1

You are given the following “**University**” class:

```
public class University{  
    public String name;  
    public String country;  
}
```

Now write a Java **tester** class named “**UniversityTester**”.

- a. Write the main method and create 2 objects of **University** class and print the location of the objects and print the instance variables of the objects. Are the location of the objects the same?
- b. Now change the instance variables of the first object.  
    name = “Imperial College London”  
    country = “England”

Now change the instance variables of the second object.  
name = “BRAC University”  
country = “Bangladesh”

Now check if the instance variables of both objects have changed or not and whether the instance variables of both objects are of the same value or not.

## Task 2

Complete the “Cat” class so the main method produces the following output:

| Test Class                                                                                                                                                                                                                                                                                                                                                                                    | Output                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <pre>public class Test7{     public static void main(String [] args){         Cat c1 = new Cat();         System.out.println("1=====");         c1.printCat();         c1.color = "Black";         System.out.println("2=====");         c1.printCat();         c1.color = "Brown";         c1.action = "jumping";         System.out.println("3=====");         c1.printCat();     } }</pre> | <pre>1===== White cat is sitting 2===== Black cat is sitting 3===== Brown cat is jumping</pre> |

## Task 3

Design the **Course** class to generate the correct output from the driver code provided below:

| Driver Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Output                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>public class Tester3{     public static void main(String[] args) {         Course c1 = new Course();         Course c2 = new Course();         c1.updateDetails("Programming Language I","CSE110", 3);         System.out.println("===== 1 =====");         c1.displayCourse();         c2.updateDetails("Data Structures","CSE220", 3);         System.out.println("===== 2 =====");         c2.displayCourse();         c1.updateDetails("Programming Language II","CSE111", 3);         System.out.println("===== 3 =====");         c1.displayCourse();     } }</pre> | <pre>===== 1 ===== Course Name: Programming Language I Course Code: CSE110 Course Credit: 3 ===== 2 ===== Course Name: Data Structures Course Code: CSE220 Course Credit: 3 ===== 3 ===== Course Name: Programming Language II Course Code: CSE111 Course Credit: 3</pre> |

## Task 4

Design the **CellPhone** class so that the **main** method of tester class can produce the following output:

| Tester Class                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Output                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>public class Tester4{     public static void main(String[]args){         CellPhone phone1 = new CellPhone();         phone1.printDetails();         phone1.model ="Nokia 1100";         System.out.println("1#####");         phone1.storeContact("Joy - 01834");         System.out.println("=====");         phone1.printDetails();         System.out.println("2#####");         phone1.storeContact("Toya - 01334");         phone1.storeContact("Aayan - 01135");         System.out.println("=====");         phone1.printDetails();         System.out.println("3#####");         phone1.storeContact("Sani - 01441");         System.out.println("=====");         phone1.printDetails();     } }</pre> | <pre>Phone Model unknown Contacts Stored 0 1##### Contact Stored ===== Phone Model Nokia 1100 Contacts Stored 1 Stored Contacts: Joy - 01834 2##### Contact Stored Contact Stored ===== Phone Model Nokia 1100 Contacts Stored 3 Stored Contacts: Joy - 01834 Toya - 01334 Aayan - 01135 3##### Memory full. New contact can't be stored. ===== Phone Model Nokia 1100 Contacts Stored 3 Stored Contacts: Joy - 01834 Toya - 01334 Aayan - 01135</pre> |

## Task 5

|    |                                                                  |
|----|------------------------------------------------------------------|
| 1  | <code>public class Task5 {</code>                                |
| 2  | <code>    public int p = 3, y = 2, sum;</code>                   |
| 3  | <code>    public void methodA(){</code>                          |
| 4  | <code>        int x = 0, y = 0;</code>                           |
| 5  | <code>        y = y + this.y;</code>                             |
| 6  | <code>        x = sum + 2 + p;</code>                            |
| 7  | <code>        sum = x + methodB(p, y) + y;</code>                |
| 8  | <code>        System.out.println(x + " " + y+ " " + sum);</code> |
| 9  | <code>    }</code>                                               |
| 10 | <code>    public int methodB(int p, int n){</code>               |
| 11 | <code>        int x = 0;</code>                                  |
| 12 | <code>        y = y + (++p);</code>                              |
| 13 | <code>        x = x + 2 + n;</code>                              |
| 14 | <code>        sum = sum + x + y;</code>                          |
| 15 | <code>        System.out.println(x + " " + y+ " " + sum);</code> |
| 16 | <code>        return sum;</code>                                 |
| 17 | <code>    }</code>                                               |
| 18 | <code>}</code>                                                   |

### Driver code:

| <pre> public class Tester5 {     public static void main(String [] args){         Task5 t1 = new Task5();         t1.methodA();         t1.methodA();     } } </pre> | Outputs |   |     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---|-----|
|                                                                                                                                                                      | x       | y | Sum |
|                                                                                                                                                                      |         |   |     |
|                                                                                                                                                                      |         |   |     |
|                                                                                                                                                                      |         |   |     |
|                                                                                                                                                                      |         |   |     |

## Task 6

|    |                                             |
|----|---------------------------------------------|
| 1  | public class Test6{                         |
| 2  | public int sum;                             |
| 3  | public int y;                               |
| 4  | public void methodA(){                      |
| 5  | int x=2, y =3;                              |
| 6  | int [] msg = {3, 7};                        |
| 7  | y = this.y + msg[0];                        |
| 8  | methodB(msg[1]++, msg[0]);                  |
| 9  | x = x + this.y + msg[1];                    |
| 10 | sum = x + y + msg[0];                       |
| 11 | System.out.println(x + " " + y+ " " + sum); |
| 12 | }                                           |
| 13 | public void methodB(int mg2, int mg1){      |
| 14 | int x = 0;                                  |
| 15 | y = this.y + mg2;                           |
| 16 | x = x + 19 + mg1;                           |
| 17 | sum = this.sum + x + y;                     |
| 18 | mg2 = y + mg1;                              |
| 19 | mg1 = mg2 + x + 2;                          |
| 20 | System.out.println(x + " " + y+ " " + sum); |
| 21 | }                                           |
| 22 | }                                           |

**Driver code:**

|                                                                                                                                                                                                         |                |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--|--|
| <pre> public class Tester6{     public static void main (String args[]){         Test6 t1 = new Test6();         t1.methodB(5,-8);         Test6 t2 = new Test6();         t2.methodA();     } } </pre> | <b>Outputs</b> |  |  |
|                                                                                                                                                                                                         |                |  |  |
|                                                                                                                                                                                                         |                |  |  |
|                                                                                                                                                                                                         |                |  |  |

# HOMEWORK

## Task 1

Design the “**ImaginaryNumber**” class to generate the **output** given below:

| Tester Class                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Output                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <pre>public class Tester7{     public static void main(String [] args){         ImaginaryNumber num1 = new ImaginaryNumber();         String p = num1.printNumber();         System.out.println(p);         System.out.println("1*****");         num1.realPart=3;         num1.imaginaryPart=7;         System.out.println(num1.printNumber());         System.out.println("2*****");         ImaginaryNumber num2 = new ImaginaryNumber();         num2.realPart=1;         num2.imaginaryPart=9;         System.out.println(num2.printNumber());     } }</pre> | <pre>0 + 0i 1***** 3 + 7i 2***** 1 + 9i</pre> |

## Task 2

Implement the “**Assignment**” class with necessary properties, so that the given output is produced for the provided driver code.

| Driver Class                                                                                                                                                                                                                                                                                                                                                              | Output                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>public class AssignmentTester{     public static void main(String [] args){         Assignment as1 = new Assignment();         as1.printDetails();         System.out.println("1-----");         as1.tasks = 11;         as1.difficulty = "Moderate";         as1.submission = true;         as1.printDetails();         System.out.println("2-----");     } }</pre> | <pre>Number of tasks: 0 Difficulty level: null Submission required: false 1----- Number of tasks: 11 Difficulty level: Moderate Submission required: true 2----- Assignment will not require submission</pre> |

|                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> System.out.println(as1.makeOptional()); System.out.println("3-----"); as1.printDetails(); System.out.println("4-----"); Assignment as2 = new Assignment(); as2.tasks = 12; as2.difficulty = "Hard"; as2.submission = false; as2.printDetails(); System.out.println("5-----"); System.out.println(as2.makeOptional());     } } </pre> | <pre> 3----- Number of tasks: 11 Difficulty level: Moderate Submission required: false 4----- Number of tasks: 12 Difficulty level: Hard Submission required: false 5----- Submission is already not required </pre> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Task 3

Create an **Employee** class to provide the expected output.

- An employee will have a name, salary and designation.
- The name will be assigned inside the newEmployee() method
- Whenever a New Employee joins his/her salary will be **Tk. 30,000** and the designation will be **junior**.
- Employees with salaries greater than **Tk. 50,000** and **Tk. 30,000** need to pay **30%** and **10%** of salary as tax respectively.
- Employees can be promoted to **senior**, **lead** and **manager** positions. Based on their promotion they will get an increment of **Tk. 25,000**, **Tk. 50,000** and **Tk. 75,000** respectively.

| Driver Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Expected Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> public class Tester9{     public static void main(String[] args){          Employee emp1 = new Employee();         Employee emp2 = new Employee();         Employee emp3 = new Employee();          emp1.newEmployee("Harry Potter");         emp2.newEmployee("Hermione Granger");         emp3.newEmployee("Ron Weasley");         System.out.println("1 =====");         emp1.displayInfo();         System.out.println("2 =====");         emp2.displayInfo();         System.out.println("3 =====");         emp3.displayInfo();         System.out.println("4 =====");         emp1.calculateTax();         System.out.println("5 =====");         emp1.promoteEmployee("lead");         System.out.println("6 =====");         emp1.calculateTax();         System.out.println("7 =====");         emp1.displayInfo();         System.out.println("8 =====");         emp3.promoteEmployee("manager");         System.out.println("9 =====");         emp3.calculateTax();         System.out.println("10 =====");         emp3.displayInfo();     } } </pre> | <pre> 1 ===== Employee Name: Harry Potter Employee Salary: 30000.0 Tk Employee Designation: junior 2 ===== Employee Name: Hermione Granger Employee Salary: 30000.0 Tk Employee Designation: junior 3 ===== Employee Name: Ron Weasley Employee Salary: 30000.0 Tk Employee Designation: junior 4 ===== No need to pay tax 5 ===== Harry Potter has been promoted to lead New Salary: 80000.00 Tk 6 ===== Harry Potter Tax Amount: 24000.0 Tk 7 ===== Employee Name: Harry Potter Employee Salary: 80000.0 Tk Employee Designation: lead 8 ===== Ron Weasley has been promoted to manager New Salary: 105000.00 Tk 9 ===== Ron Weasley Tax Amount: 31500.0 Tk 10 ===== Employee Name: Ron Weasley Employee Salary: 105000.0 Tk Employee Designation: manager </pre> |

## Task 4

Implement the “**MobilePhone**” class with necessary properties to produce the given output for the provided driver code.

| Driver Class                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>public class MobilePhoneTester{     public static void main(String args []){         MobilePhone m1 = new MobilePhone();         MobilePhone m2 = new MobilePhone();         m1.setContactCapacity(5);         m2.setContactCapacity(100);         m1.details();         System.out.println("1-----");         m1.addContact("John", 9866);         m1.addContact("Maria", 7865);         System.out.println("2-----");         m1.details();         System.out.println("3-----");         m1.makeCall(9866);         System.out.println("4-----");         m1.addContact("Henry", 2365);         System.out.println("5-----");         m1.makeCall(7552);         m1.makeCall(2365);         System.out.println("6-----");         m1.addContact("Gomes", 4589);         m1.addContact("Antony", 8421);         m1.addContact("Tony", 5789);         System.out.println("7-----");         m1.details();     } }</pre> | <pre>Total Contacts: 0 Contact List: 1----- The contact of John is added. The contact of Maria is added. 2----- Total Contacts: 2 Contact List: John:9866 Maria:7865 3----- Calling John . . . 4----- The contact of Henry is added. 5----- Calling 7552 . . . Calling Henry . . . 6----- The contact of Gomes is added. The contact of Antony is added. Storage Full!! 7----- Total Contacts: 5 Contact List: John:9866 Maria:7865 Henry:2365 Gomes:4589 Antony:8421</pre> |

## Task 5

|    |                                                 |
|----|-------------------------------------------------|
| 1  | public class B {                                |
| 2  | public int temp = 4;                            |
| 3  | public int sum, y, x;                           |
| 4  | public void methodA(int m){                     |
| 5  | int [] n = {2,5};                               |
| 6  | int x = 0;                                      |
| 7  | y = m + this.methodB(x++,m)+(temp++);           |
| 8  | x = this.x + 2 + n[0];                          |
| 9  | sum = sum + x + y;                              |
| 10 | n[0] = sum + 2;                                 |
| 11 | System.out.println(n[0]+" " + x+ " " + sum);    |
| 12 | }                                               |
| 13 | public int methodB(int m, int n){               |
| 14 | int y = 4 + this.y + m;                         |
| 15 | x = this.y + y + (++temp) - n;                  |
| 16 | sum = x + y + this.sum;                         |
| 17 | System.out.println(y+ " " + this.x + " " +sum); |
| 18 | return x;                                       |
| 19 | }                                               |
| 20 | }                                               |

|                                                                                                                                                                                             |                |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--|--|
| <pre> public class Tester11 {     public static void main(String [] args){         B t1 = new B();         t1.methodA(5);         B t2 = new B();         t2.methodB(12, 2);     } } </pre> | <b>Outputs</b> |  |  |
|                                                                                                                                                                                             |                |  |  |
|                                                                                                                                                                                             |                |  |  |
|                                                                                                                                                                                             |                |  |  |

## Task 6

|    |                                                   |
|----|---------------------------------------------------|
| 1  | public class A{                                   |
| 2  | public int x = 3, y = 5, sum = 9;                 |
| 3  | public int methodA(int temp, int x){              |
| 4  | this.x += (x++) + temp;                           |
| 5  | if(temp % 5 == 2){                                |
| 6  | sum += (this.y++) + temp;                         |
| 7  | }                                                 |
| 8  | else{                                             |
| 9  | sum += 3;                                         |
| 10 | if(y > 5) ++y;                                    |
| 11 | }                                                 |
| 12 | System.out.println(this.x + " " + y + " " + sum); |
| 13 | return this.x;                                    |
| 14 | }                                                 |
| 15 | public void methodB(int y){                       |
| 16 | int temp = (y++) + this.y;                        |
| 17 | this.y = (++temp) + methodA(temp, y) + x;         |
| 18 | sum = y + (++this.x) + (temp++);                  |
| 19 | System.out.println(x + " " + y + " " + (sum++));  |
| 20 | }                                                 |
| 21 | public void methodC(int y){                       |
| 22 | y = (this.x++) + sum + 3;                         |
| 23 | System.out.println(x + " " + y + " " + sum);      |
| 24 | }                                                 |
| 25 | }                                                 |

### Driver code:

| <pre> public class Test12{     public static void main(String [] args) {         A a1 = new A();         a1.methodA(2, 4);         a1.methodB(3);         A a2 = new A();         a2.methodC(7);         a1.methodB(3);     } } </pre> | Outputs |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--|--|
|                                                                                                                                                                                                                                        |         |  |  |
|                                                                                                                                                                                                                                        |         |  |  |
|                                                                                                                                                                                                                                        |         |  |  |
|                                                                                                                                                                                                                                        |         |  |  |

## Ungraded Tasks (Optional)

(You don't have to submit the ungraded tasks)

### Task 1

Complete the **Bird** class so that main method produces the following **output**:

| Test class                                                                                                                                                                                                                                                                                                                                                                                      | Output                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>public class BirdTest{     public static void main(String args[]) {         Bird b1 = new Bird();         b1.name = "Parrot";         b1.flyUp(3);         b1.makeNoise();         b1.flyDown(5);         b1.flyDown(2);         b1.flyDown(1);         Bird b2 = new Bird();         b2.name = "Eagle";         b2.flyUp(5);         b2.flyDown(5);         b2.makeNoise();     } }</pre> | <pre>Parrot has flown up 3 feet. Squawk Parrot cannot fly down 5 feet. Parrot has flown down 2 feet. Parrot has flown down 1 feet and landed. Eagle has flown up 5 feet. Eagle has flown down 5 feet and landed. Squee</pre> |

### Task 2

Implement the “**ChickenBurger**” class with necessary properties, so that the given output is produced for the provided driver code.

[Note:

1. There are four available spice levels: **Mild**, **Spicy**, **Naga** and **Extreme**. You can store these values in a String array.
2. You might need to use the `.equals()` method to compare two string values.]

| Driver Class                                                                                                                                                                  | Output                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <pre>public class BurgerMaker{     public static void main(String [] args){         ChickenBurger b1 = new ChickenBurger();         System.out.println(b1.bun);     } }</pre> | <pre>Sesame 200 Less Not Set -----1----- Cannot serve now. Customize Spice</pre> |

```

System.out.println(b1.price);
System.out.println(b1.sauceOption);
System.out.println(b1.spiceLevel);
System.out.println("-----1-----");
System.out.println(b1.serveBurger());
System.out.println("-----2-----");
b1.customizeSpiceLevel("Extreme Jhaal");
b1.customizeSpiceLevel("Spicy");
System.out.println("-----3-----");
System.out.println(b1.serveBurger());
System.out.println("-----4-----");
ChickenBurger b2 = new ChickenBurger();
b2.bun = "Brioche";
b2.price += 50;
b2.sauceOption = "Regular";
b2.customizeSpiceLevel("Naga");
System.out.println("-----5-----");
System.out.println(b2.serveBurger());
}
}

```

```

Level first.
-----2-----
This spice level is unavailable.
Spice level set to Spicy.
-----3-----
The burger is being served:-
Bun Type: Sesame
Price: 200
Sauce Option: Less
Spice Level: Spicy
-----4-----
Spice level set to Naga.
-----5-----
The burger is being served:-
Bun Type: Brioche
Price: 250
Sauce Option: Regular
Spice Level: Naga

```

### Task 3

Design the **MagicPotion** class so that the following Tester class generates the expected output

- Potion cannot be activated if potion strength below 5

| Driver Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Outputs                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>public class MagicTester {     public static void main(String[] args) {         MagicPotion potion = new MagicPotion();         potion.setDetails("blue", 4);         System.out.println("=====");         potion.activate();         potion.describe();         System.out.println("=====");         potion.boost();         potion.describe();         System.out.println("=====");         potion.activate();         potion.describe();         System.out.println("=====");         System.out.println("Is potion strong? " +             potion.isStrong());         System.out.println("=====");         potion.boost();         potion.activate();         potion.describe();         System.out.println("=====");         System.out.println("Is potion strong? " +             potion.isStrong());         System.out.println("=====");         potion.boost();         System.out.println("Is potion strong? " +             potion.isStrong());         System.out.println("=====");         potion.weaken();         potion.weaken();         potion.weaken();         potion.describe();     } }</pre> | <pre>===== Potion colour: blue Potion strength: 4 Is active: false ===== Potion colour: blue Potion strength: 5 Is active: false ===== Potion colour: blue Potion strength: 5 Is active: true ===== Is potion strong? false ===== Already activated Potion colour: blue Potion strength: 6 Is active: true ===== Is potion strong? false ===== Is potion strong? true ===== Potion colour: blue Potion strength: 4 Is active: false</pre> |

### Task 4

|    |                                                       |
|----|-------------------------------------------------------|
| 1  | public class TraceA{                                  |
| 2  | public int x =0, y = 8, sum = 0;                      |
| 3  | public void methodA(int y, int x){                    |
| 4  | int [] arr = {4, 7};                                  |
| 5  | this.x += x;                                          |
| 6  | arr[0] = y++;                                         |
| 7  | arr[1] += arr[1] % arr[0] * x;                        |
| 8  | if ( y % 2 == 0){                                     |
| 9  | sum = x + methodB(arr[0]++, arr[1], y) + this.x;      |
| 10 | }                                                     |
| 11 | else{                                                 |
| 12 | sum = this.y + methodB(++arr[0], arr[1], x) + this.y; |
| 13 | }                                                     |
| 14 | System.out.println(x+ " " + arr[0] + " " + sum);      |
| 15 | }                                                     |
| 16 | public int methodB(int a, int b, int x){              |
| 17 | sum = sum % x;                                        |
| 18 | if( a % b == x ){                                     |
| 19 | return this.y--;                                      |
| 20 | }                                                     |
| 21 | this.x = a * b / x;                                   |
| 22 | y += this.x % this.y;                                 |
| 23 | System.out.println(a + " " + this.x + " " + y);       |
| 24 | return y;                                             |
| 25 | }                                                     |
| 26 | }                                                     |

#### Driver code:

| <pre> public class TesterX{     public static void main(String [] args){         TraceA t1 = new TraceA();         t1.methodA(4, 7);         int x = t1.methodB(3,2,1);         System.out.println(t1.y + " " + t1.sum + " " + x);     } } </pre> | Output |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--|--|
|                                                                                                                                                                                                                                                   |        |  |  |
|                                                                                                                                                                                                                                                   |        |  |  |
|                                                                                                                                                                                                                                                   |        |  |  |